Installing the Jetpack Environment

Installing the Jetpack Environment
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Some users may need to use NVIDIA's built-in system component environment. After successfully installing the system and entering the desktop, you can follow the tutorial below to install the component environment! Since the Jetpack environment is relatively large, it is recommended to install it on an NVMe SSD system environment.

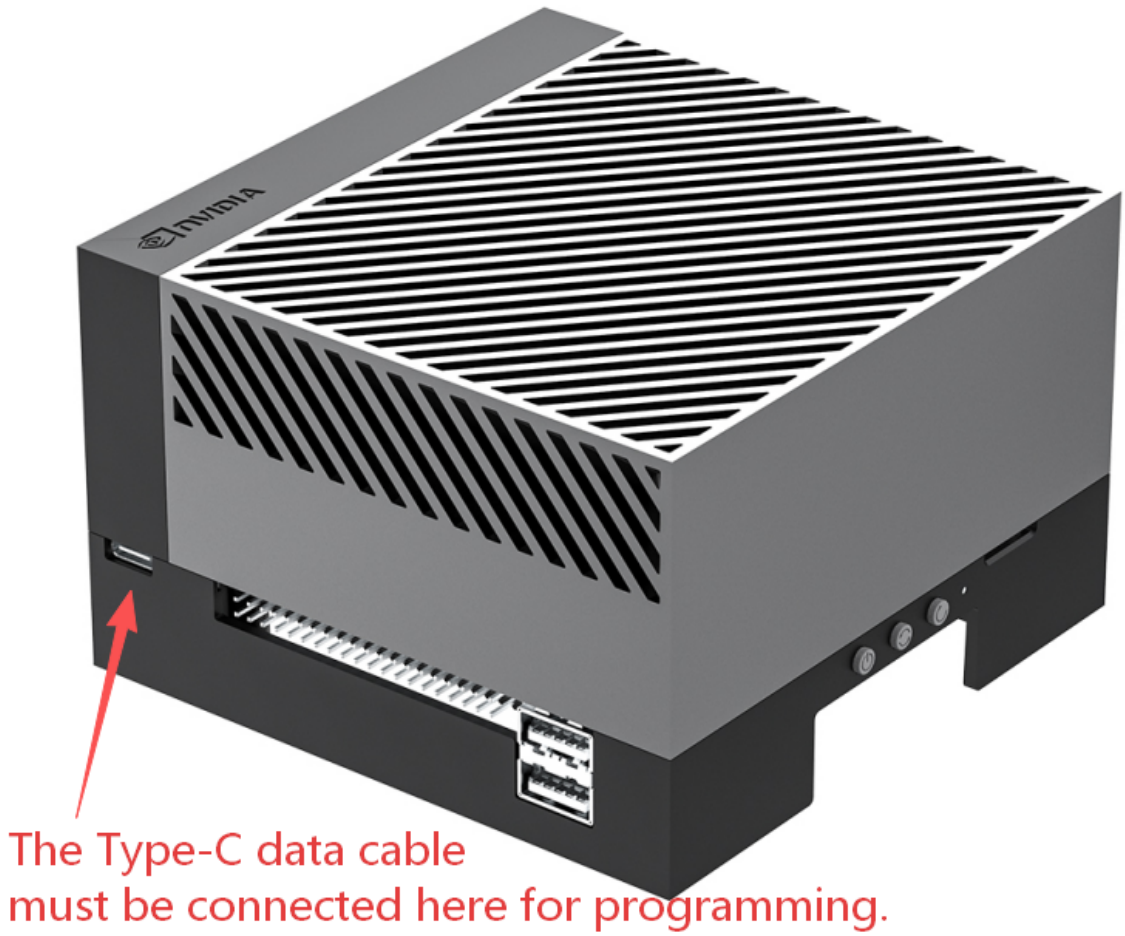
Note: The SSD included with the product contains a factory image, which includes the official NVIDIA system and motherboard environment; unless a clean system is required, please do not proceed!

```
The entire process requires a network connection. Some components may require a
VPN. If you cannot complete the component installation, use our factory image.
```

1. Hardware Connection

Plug the power adapter into the Jetson AGX Orin motherboard. Connect the Type-C data cable to your Ubuntu computer or virtual machine. The other end must be connected to the Type-C port next to the 40-pin header on the Jetson AGX Orin motherboard.

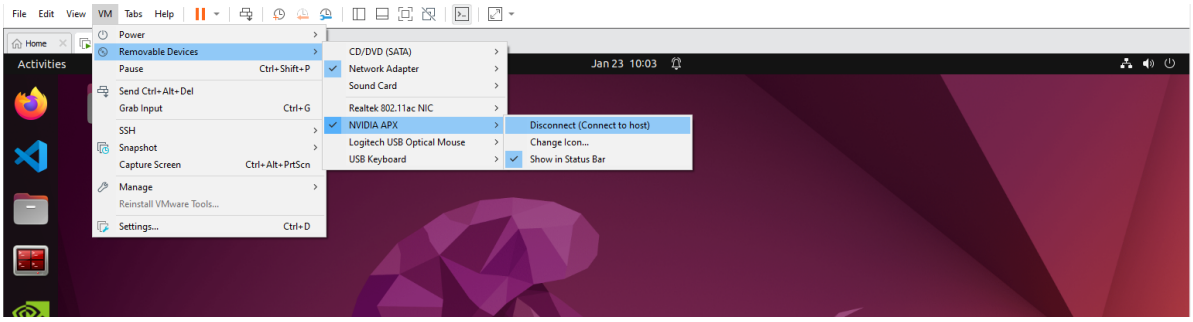
Note: Installing the Jetpack component environment does not require entering flashing mode; the motherboard only needs to be powered on!



2. Software Connection

This tutorial uses VMware Workstation 17 Pro to flash the system!

After starting the virtual machine, select in the menu bar: Virtual Machine → NVIDIA Linux for Tegra → Confirm the connection status with the virtual machine.



3. Using SDK Manager

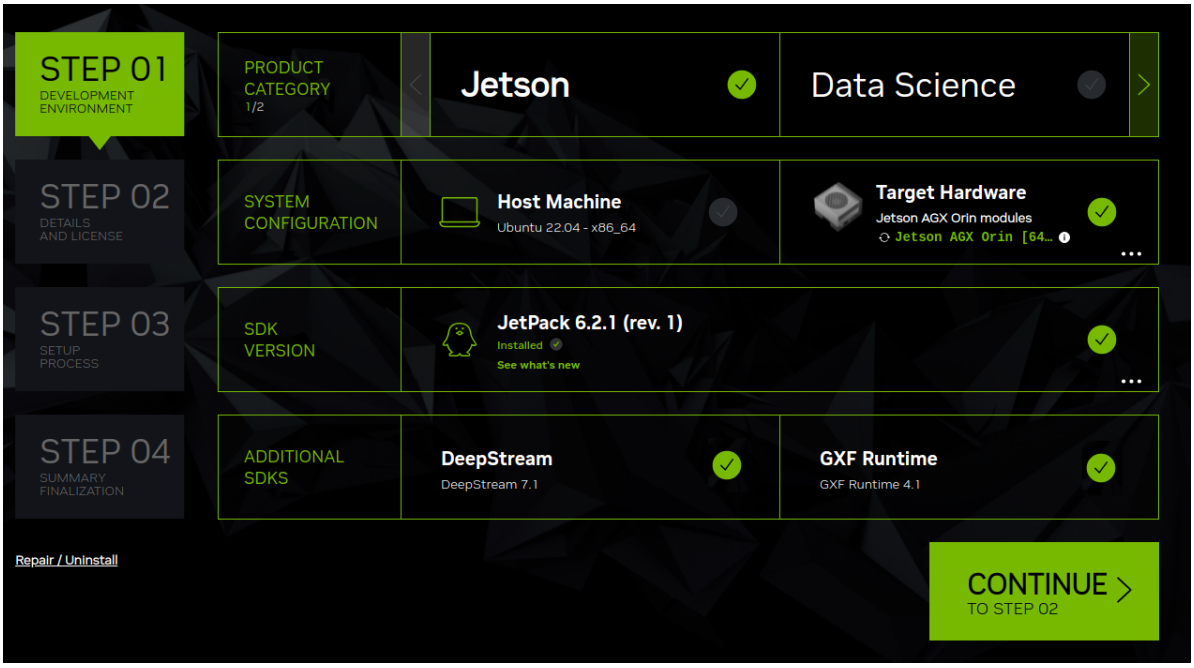
With the Jetson AGX Orin motherboard in normal system startup mode, connect the Ubuntu computer or virtual machine using a Type-C data cable.

3.1 Motherboard Selection

After opening `SDK Manager`, select according to your motherboard model: For official packages, select `Jetson AGX Orin [64GB developer kit version]`

3.2 STEP 1

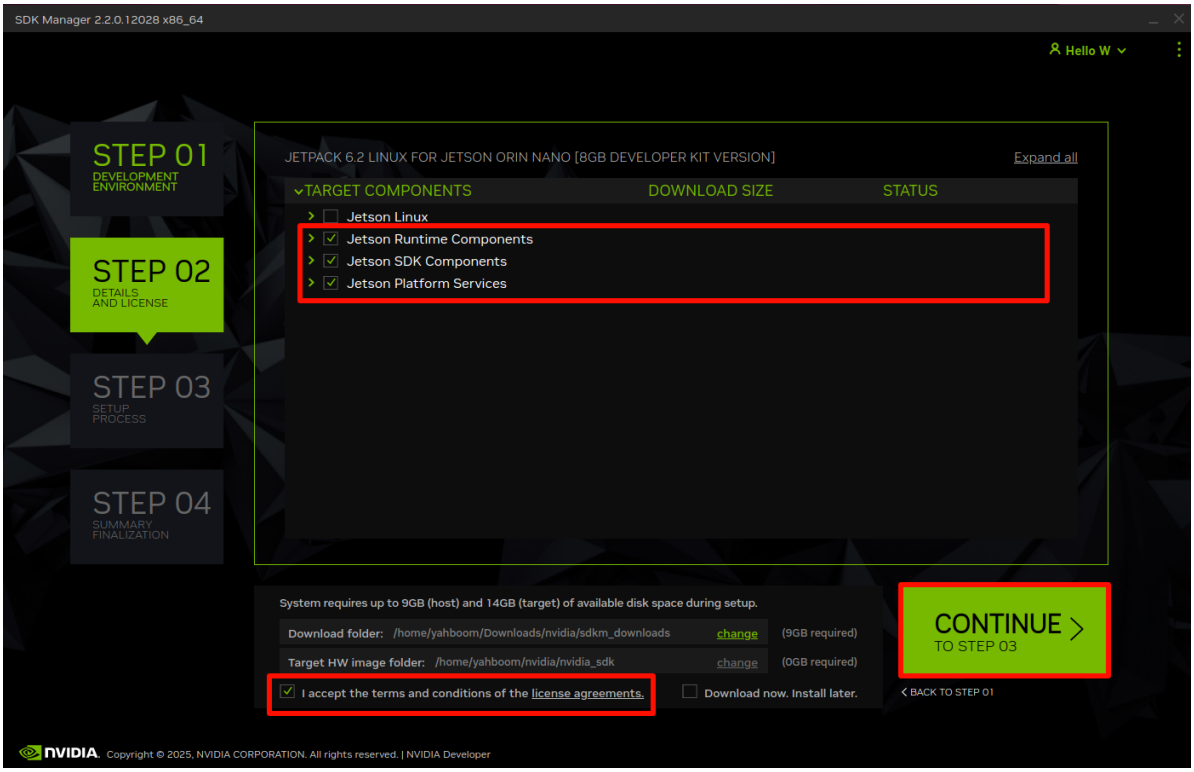
Check the options below, and after confirming that all selections are correct, click `CONTINUE`:



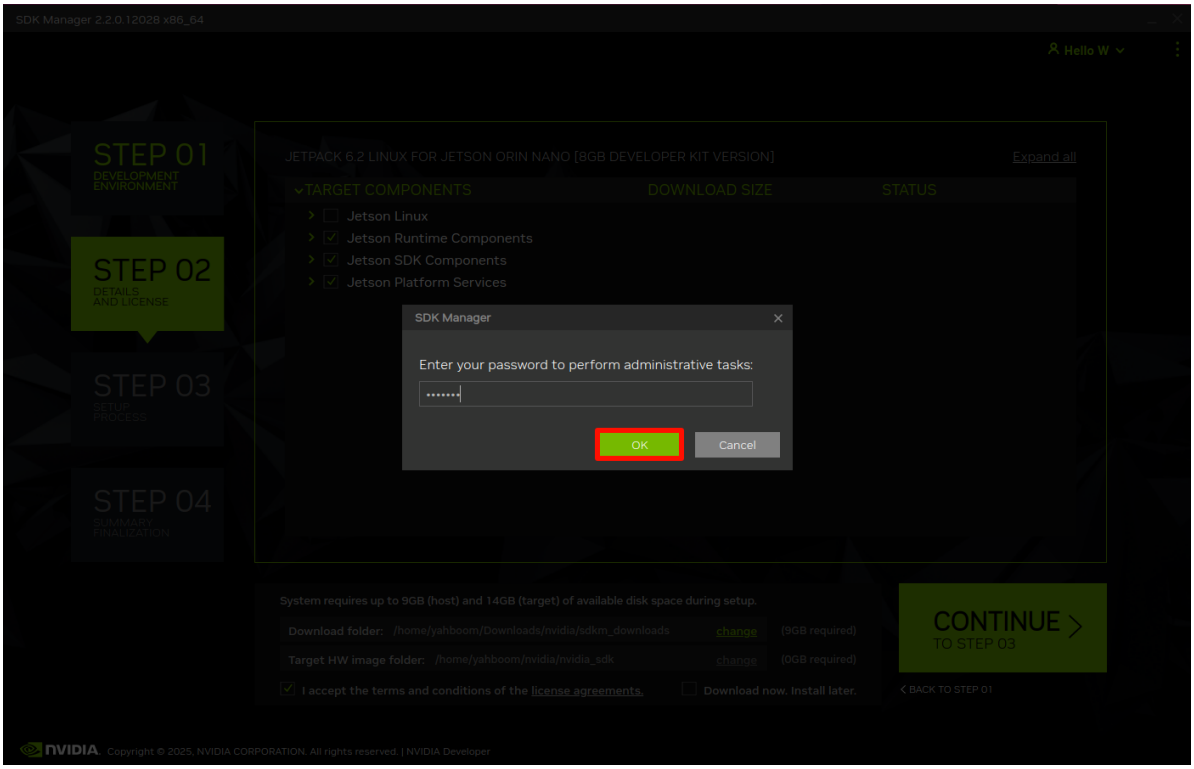
3.3 STEP 2

Note: The system has already been successfully installed; do not select the Jetson Linux option.

We only need to select the required component environments: SDK Manager will automatically download all components. Once the download is complete, we can proceed to the next step.

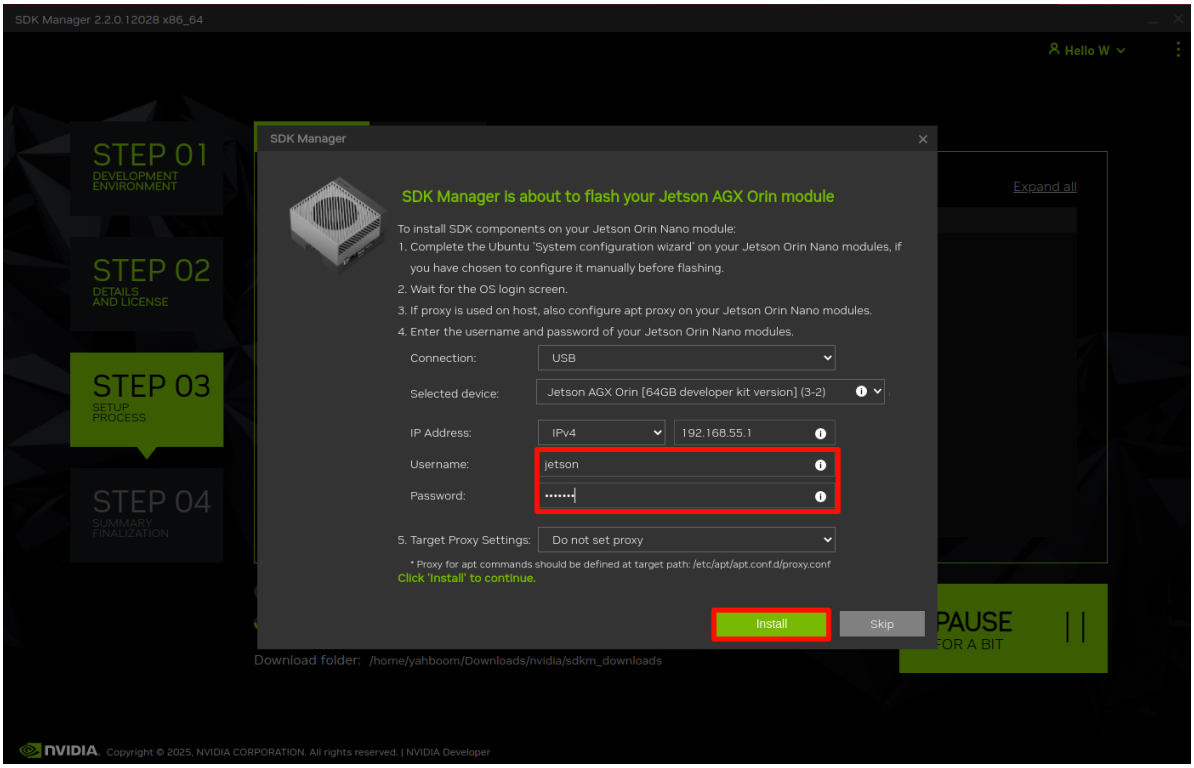


Enter the virtual machine password: yahboom

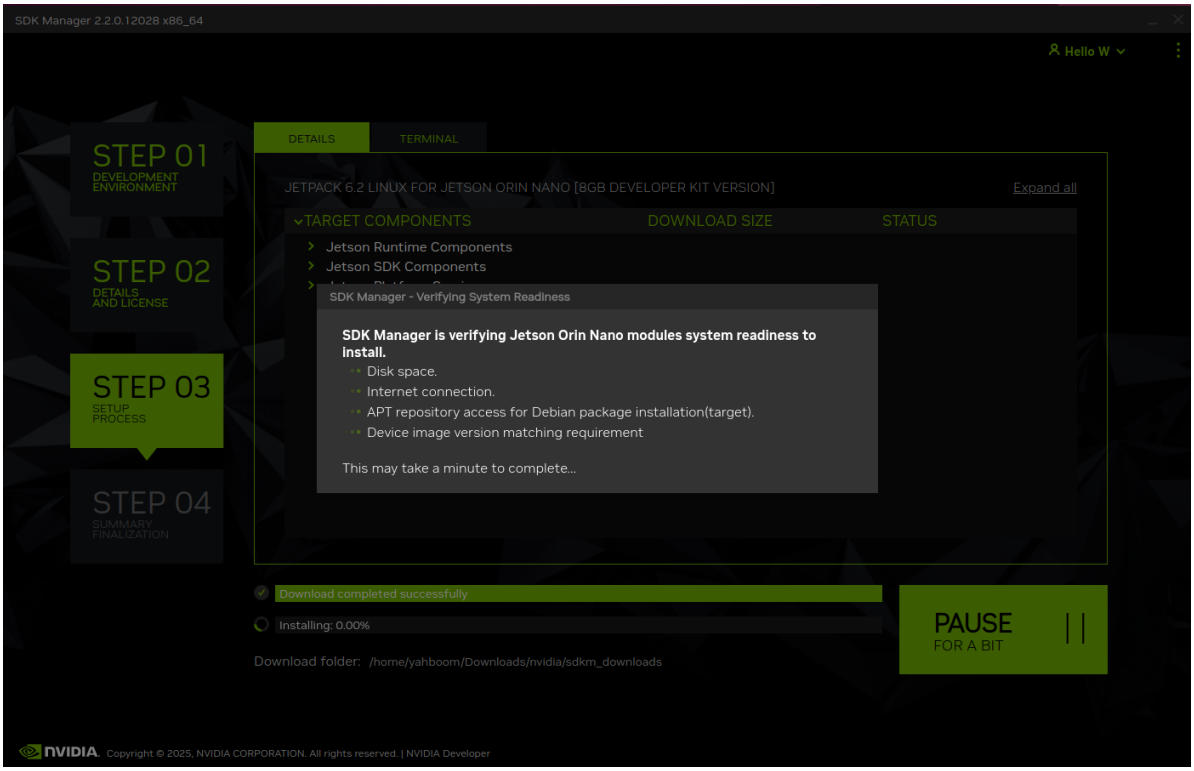


3.4 STEP 3

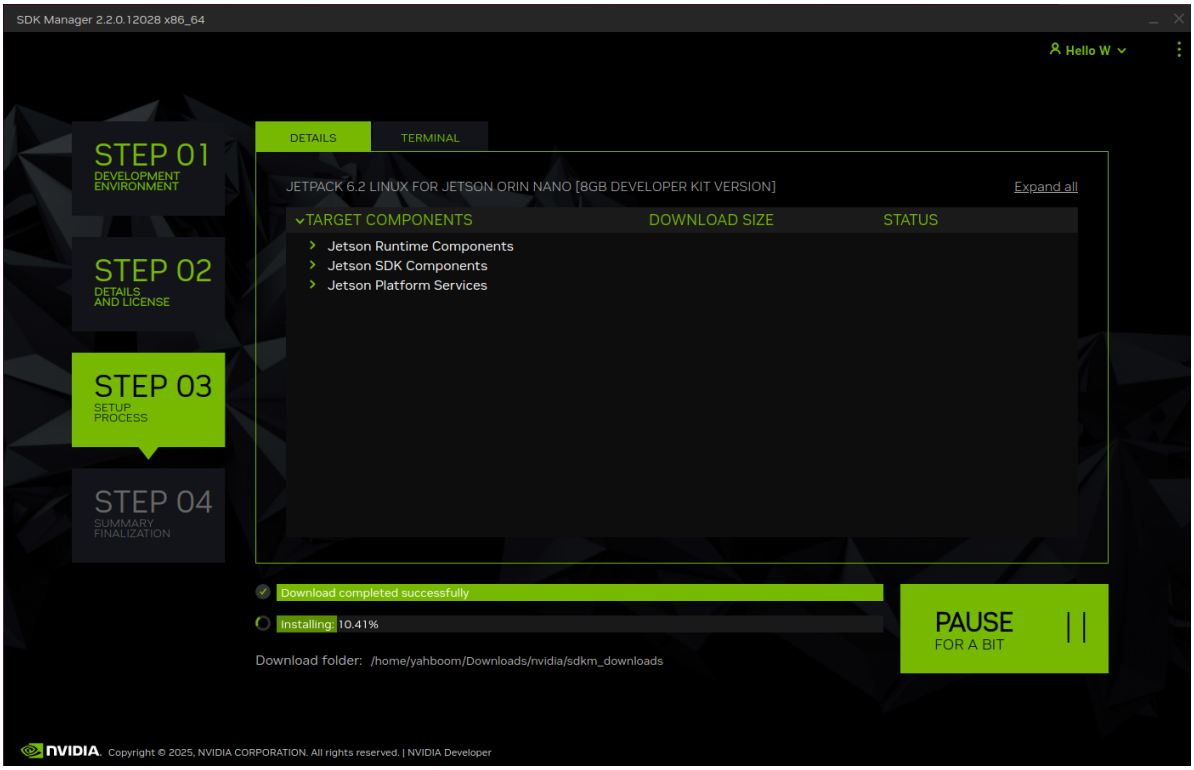
Here, enter the username and password you set during system startup. This process takes a long time, please be patient.



Wait for the system to download and install: A network failure occurred in the virtual machine during this process, which can be ignored. Testing showed it did not affect the flashing of component environments.



The entire installation process takes a long time, please be patient. If the installation fails, you can restart the motherboard system and restart the SDK. Manager Installation:


SDK Manager 2.2.0.12028 x86_64
Hello W
STEP 01
DEVELOPMENT ENVIRONMENT
STEP 02
DETAILS AND LICENSE
STEP 03
SETUP PROCESS
STEP 04
SUMMARY FINALIZATION
DETAILS
TERMINAL
JETPACK 6.2 LINUX FOR JETSON ORIN NANO [8GB DEVELOPER KIT VERSION]
Expand all
TARGET COMPONENTS
DOWNLOAD SIZE
STATUS
Jetson Runtime Components
Jetson SDK Components
Jetson Platform Services
Download completed successfully
Installing: 10.41%
Download folder: /home/yahboom/Downloads/nvidia/sdkm_downloads
PAUSE
FOR A BIT
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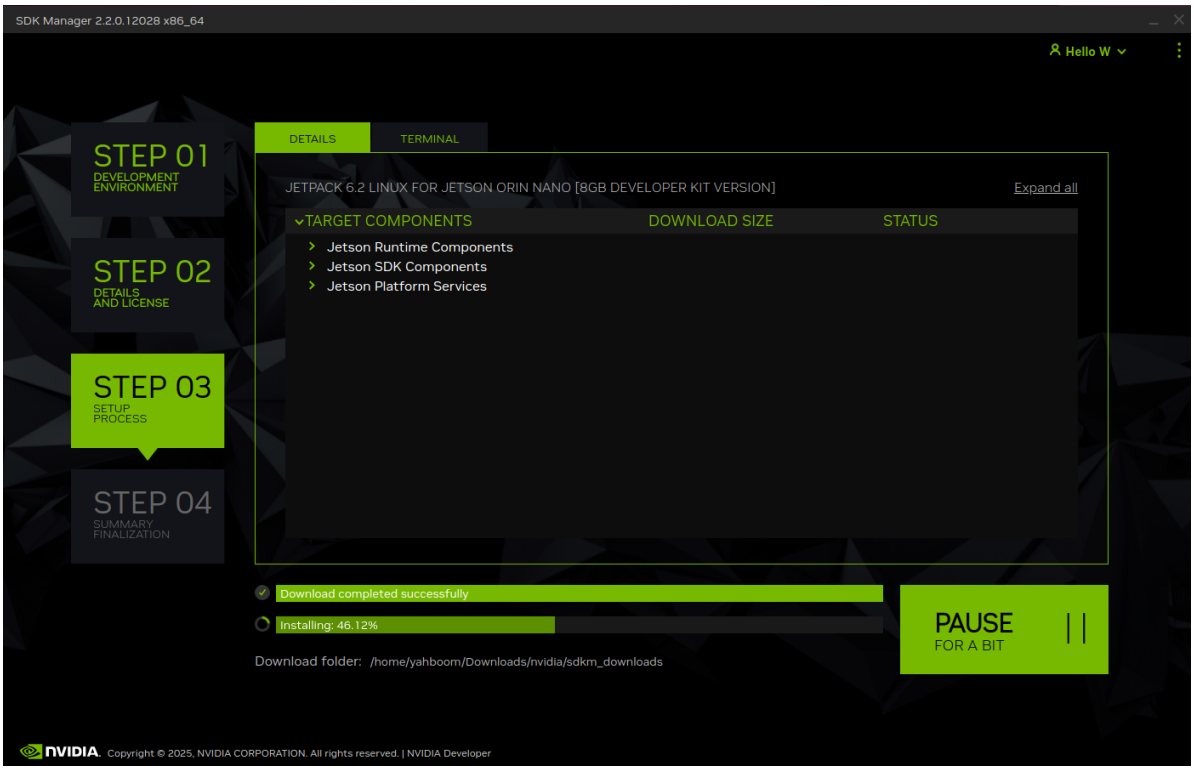
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Expand all
TARGET COMPONENTS
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STATUS
Jetson Runtime Components
Jetson SDK Components
Jetson Platform Services
Download completed successfully
Installing: 46.12%
Download folder: /home/yahboom/Downloads/nvidia/sdkm_downloads
PAUSE
FOR A BIT
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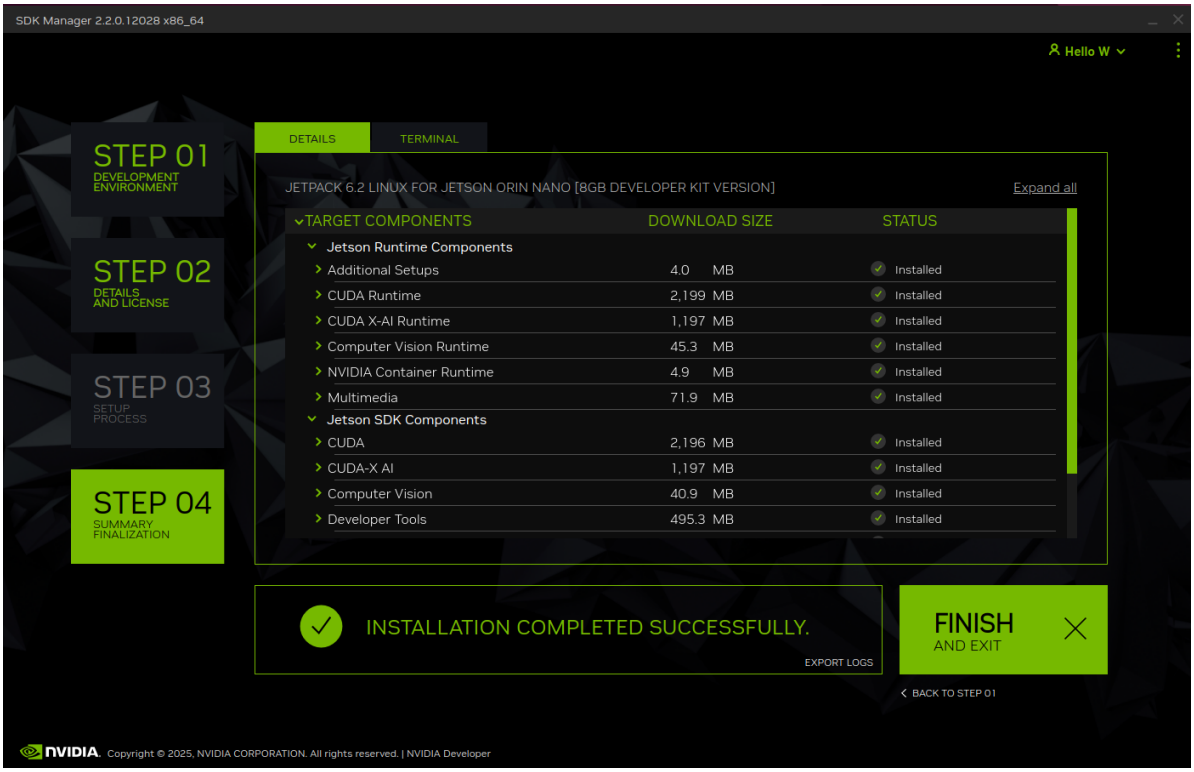
3.5, STEP4



4, Environment Verification

> **Jtop Tool**

Install Jtop using the following commands:

```
sudo apt update && sudo apt upgrade
sudo apt install python3-pip -y
sudo pip3 install -U jetson-stats
```

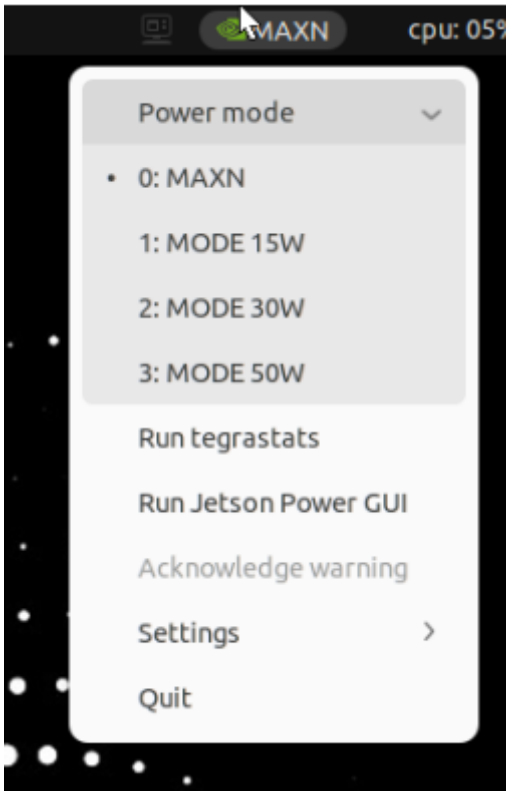
Note: A system restart is required after installation!

Optimal Performance Mode

Enable MAX Power Mode

```
sudo nvpmodel -m 0
```

Alternatively, find the NVIDIA icon in the upper right corner of your desktop and change the power mode to MAXN.

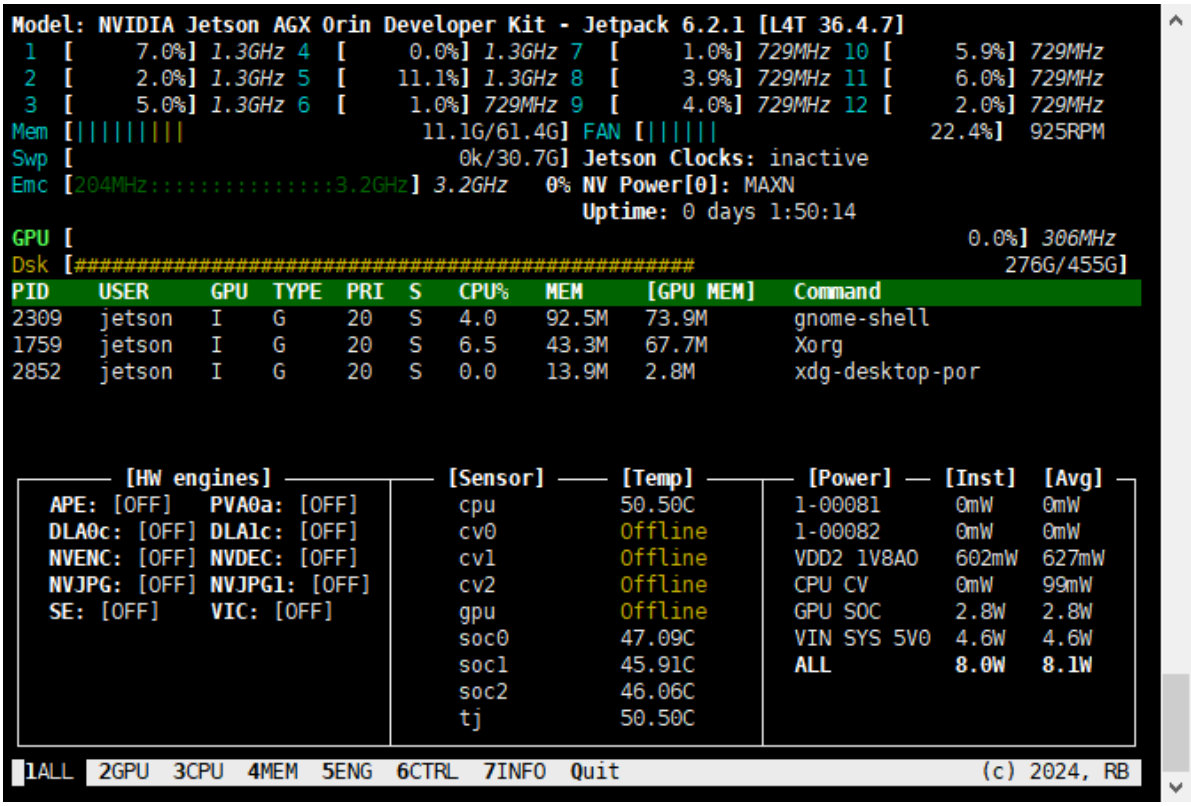


Enable Jetson clocks: CPU and GPU cores run at maximum frequency

```
sudo jetson_clocks
```

Use the Jtop tool to view system information:

```
jtop
```



```
jtop 4.3.2 - (c) 2024, Raffaello Bonghi [raffaello@rnext.it]
Website: https://rnext.it/jetson_stats

 Platform                                        Serial Number: [s|XX CLICK TO READ XXX]
  Machine: aarch64                               Hardware
  System: Linux                                   Model: NVIDIA Jetson AGX Orin Developer Kit
  Distribution: Ubuntu 22.04 Jammy Jellyfish      699-level Part Number: 699-13701-0005-501 G.0
  Release: 5.15.148-tegra                         P-Number: p3701-0005
  Python: 3.10.12                                 Module: NVIDIA Jetson AGX Orin (64GB ram)
                                                  SoC: tegra234
 Libraries                                        CUDA Arch BIN: 8.7
  CUDA: 12.6.85                                   L4T: 36.4.7
  cuDNN: 9.18.1.3                                 Jetpack: 6.2.1
  TensorRT: 10.7.0.23
  VPI: 3.2.4                                     Hostname: ubuntu
  Vulkan: 1.3.204                                Interfaces
  OpenCV: 4.8.0 with CUDA: NO                     wlP1p1s0: 192.168.10.99
                                                  docker0: 172.17.0.1

 1ALL  2GPU  3CPU  4MEM  5ENG  6CTRL  7INFO  Quit                                (c) 2024, RB
```